

Shalom Lakew

GIS Graduate Student | Environmental GIS | Hydrology | Remote Sensing

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Portfolio: <https://shalomlakew.github.io/>

GitHub: <https://github.com/shalomlakew/>

PROFESSIONAL SUMMARY

Graduate student pursuing a Master of Science in Geographic Information Systems (GIS) at the University of Cincinnati with a Bachelor of Science in Earth and Environmental Science. Experienced in GIS-based spatial analysis, environmental mapping, remote sensing interpretation, watershed analysis, and geographic data visualization. Skilled in applying GIS methods to analyze environmental systems, land use patterns, and hydrologic processes. Interested in environmental GIS, water resources, and spatial data analysis.

EDUCATION

University of Cincinnati

Master of Science in Geographic Information Systems (GIS)

Currently Enrolled

Relevant Coursework:

- GIS in Hydrology
- Fundamentals of GIS Programming
- Spatial Analysis
- Remote Sensing
- Advanced GIS Applications

University of Illinois Chicago

Bachelor of Science in Earth and Environmental Science, Cum Laude

August 2021 – December 2024

GPA: 3.7/4.0

Honors:

- Honors and High Distinction in Earth and Environmental Science

Relevant Coursework:

- Introduction to GIS
- Analytical Cartography
- Intermediate GIS
- Remote Sensing
- Quantitative Geospatial Statistics
- GIS and Urban Geography
- GIS in Hydrology
- Earth Systems
- Soils and the Environment
- Sedimentary Environments
- Field Work in Missouri
- Independent Research

Harper College

Associate Degree in General Science

August 2019 – July 2021

GPA: 3.8/4.0

TECHNICAL SKILLS

GIS Software

- ArcGIS Pro

- ArcGIS Online
- ArcGIS StoryMaps
- QGIS

GIS Analysis

- Spatial analysis
- Raster and vector analysis
- Digital Elevation Model (DEM) processing
- Watershed analysis
- Hydrologic modeling
- Land cover analysis
- Remote sensing interpretation
- Spatial interpolation
- Zonal statistics
- Buffer analysis
- Spatial joins
- Geodatabase development
- Cartographic design

Data & Programming

- GIS data management
- Microsoft Excel
- Scientific data analysis
- Python GIS programming (in progress)

Research Skills

- Environmental research methods
- Scientific writing
- Data interpretation
- Experimental design
- Technical reporting

Languages

- Amharic
- English

GIS PROJECTS

Watershed Terrain and Precipitation Analysis

San Marcos Basin, Texas

- Applied DEM-based spatial analysis techniques to calculate slope, aspect, flow direction, and hydrologic slope.
- Developed an ArcGIS ModelBuilder workflow to automate terrain processing and improve GIS workflow efficiency.
- Used zonal statistics, contour mapping, hillshade analysis, Thiessen polygons, and spline interpolation to evaluate elevation and precipitation patterns.
- Analyzed relationships between terrain characteristics, drainage patterns, and watershed processes.

Skills: ArcGIS Pro | Raster Analysis | ModelBuilder | Hydrology GIS | Spatial Statistics

Watershed Database Development and Hydrologic Mapping

San Marcos Basin, Texas

- Developed a GIS geodatabase integrating watershed boundaries, subwatersheds, stream networks, catchments, land cover data, and USGS stream gauge locations.
- Processed NLCD land cover data to evaluate environmental characteristics influencing watershed behavior.
- Created professional cartographic products showing watershed structure, drainage networks, and monitoring infrastructure.
- Applied GIS data organization and visualization techniques to support hydrologic analysis.

Skills: Geodatabase Management | Hydrologic Data | Cartography | GIS Data Integration

Thematic Mapping and Urban Accessibility Analysis Using GIS

- Created thematic maps analyzing global population distribution, U.S. demographic patterns, and urban park accessibility.
- Applied projection selection, definition queries, attribute joins, choropleth mapping, and graduated symbol techniques.
- Conducted buffer analysis to evaluate walkable access to public parks in Cincinnati.

- Designed professional map layouts following cartographic principles.

Skills: Thematic Cartography | Urban GIS | Spatial Analysis | Map Design

Urban Land Cover Analysis Using GIS and Remote Sensing

Baton Rouge, Louisiana | Buffalo, New York | Omaha, Nebraska

- Analyzed land cover patterns to compare urban development, forests, wetlands, agriculture, and industrial areas across three U.S. cities.
- Evaluated relationships between land cover characteristics, population patterns, and geographic conditions.
- Created thematic maps to communicate differences in urban and environmental landscapes.
- Applied remote sensing interpretation methods to analyze land use patterns.

Skills: Remote Sensing | Land Cover Analysis | Environmental GIS | Cartography

EXPERIENCE

Subtitle Translator (Volunteer)

The Chosen | Virtual

2019 – Present

- Translated English subtitles into Amharic to improve accessibility for international audiences.
- Maintained accuracy, consistency, and attention to detail while working with technical language content.

Volunteer

Awana | Hoffman Estates, IL

2019 – 2020

- Organized and prepared materials for international distribution.
- Collaborated with team members to complete volunteer activities efficiently.

Volunteer

WINGS | Hoffman Estates, IL

2019

- Assisted customers and supported daily organizational operations.
- Maintained organized work areas and contributed to team activities.

PROFESSIONAL INTERESTS

- Environmental GIS
- Hydrologic Applications of GIS
- Remote Sensing
- Watershed Management
- Urban Environmental Analysis
- Spatial Data Visualization